



ESKISEHIR OSMANGAZI UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING
2021-2022 SPRING SEMESTER
DIGITAL SYSTEMS LABORATORY

EXPERIMENT 9
Checkup Test With Mealy Machine



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EXPERIMENT 9: Checkup Test With Mealy Machine

OBJECTIVES

Our goal in this experiment is to design an up-to-date, usable or developable circuit mealy machine using the information we see and learn in the digital Lab.

APPARATUS

- Xilinx
- 7474 Flip Flops
- And Gate, Or Gate and Clock Pulse
- Proteus

INTRODUCTION

Mealy Machine

- 1-Output depends both upon the present state and the present input
- 2-Generally, it has fewer states than Moore Machine.
- 3-The value of the output function is a function of the transitions and the changes, when the input logic on the present state is done.
- 4-Mealy machines react faster to inputs. They generally react in the same clock cycle.

PROCEDURE

- 1- Create a patient's check up list with the mealy circuit.
- 2- Create an accuracy table on paper and make the necessary simplifications.
- 3-Draw the established circuit on the proteus.
- 4-Write Xilinx code in your project.
- 5-Interpret your interpretations of the experiment in the conversation and Results section.

REFERENCES

- 1- <https://www.medicalpark.com.tr/check-up/hg-1929>
- 2- <https://www.youtube.com/watch?v=M0mx8S05v60&list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eA0m>
- 3- https://www.tutorialspoint.com/automata_theory/moore_and_mealy_machines.htm
- 4- <https://www.geeksforgeeks.org/mealy-and-moore-machines-in-toc/>

D_1	$\bar{D}\bar{E}$	$\bar{D}E$	$D\bar{E}$	DE
$\bar{A}\bar{B}\bar{C}$	0	0	0	0
$\bar{A}\bar{B}C$	0	0	0	0
$\bar{A}B\bar{C}$	0	0	1	0
$\bar{A}BC$	0	0	0	0
$AB\bar{C}$	1	1	1	1
$A\bar{B}\bar{C}$	1	1	0	1
$AB\bar{C}$	x	x	x	x
ABC	x	x	x	x

$$D_1 = AC' + AD' + AE + BCDE$$

D_2	$\bar{D}\bar{E}$	$\bar{D}E$	$D\bar{E}$	DE
$\bar{A}\bar{B}\bar{C}$	0	0	0	0
$\bar{A}\bar{B}C$	0	0	1	0
$\bar{A}B\bar{C}$	1	1	0	1
$\bar{A}BC$	1	1	1	1
$AB\bar{C}$	0	0	0	0
$A\bar{B}\bar{C}$	0	0	0	0
$AB\bar{C}$	x	x	x	x
ABC	x	x	x	x

$$D_2 = BC' + AD' + BE' + A'DCDE$$

D_3	$\bar{D}\bar{E}$	$\bar{D}E$	$D\bar{E}$	DE
$\bar{A}\bar{B}\bar{C}$	0	0	1	0
$\bar{A}\bar{B}C$	1	1	0	1
$\bar{A}B\bar{C}$	1	1	0	1
$\bar{A}BC$	0	0	1	0
$AB\bar{C}$	0	0	1	0
$A\bar{B}\bar{C}$	1	1	0	1
$AB\bar{C}$	x	x	x	x
ABC	x	x	x	x

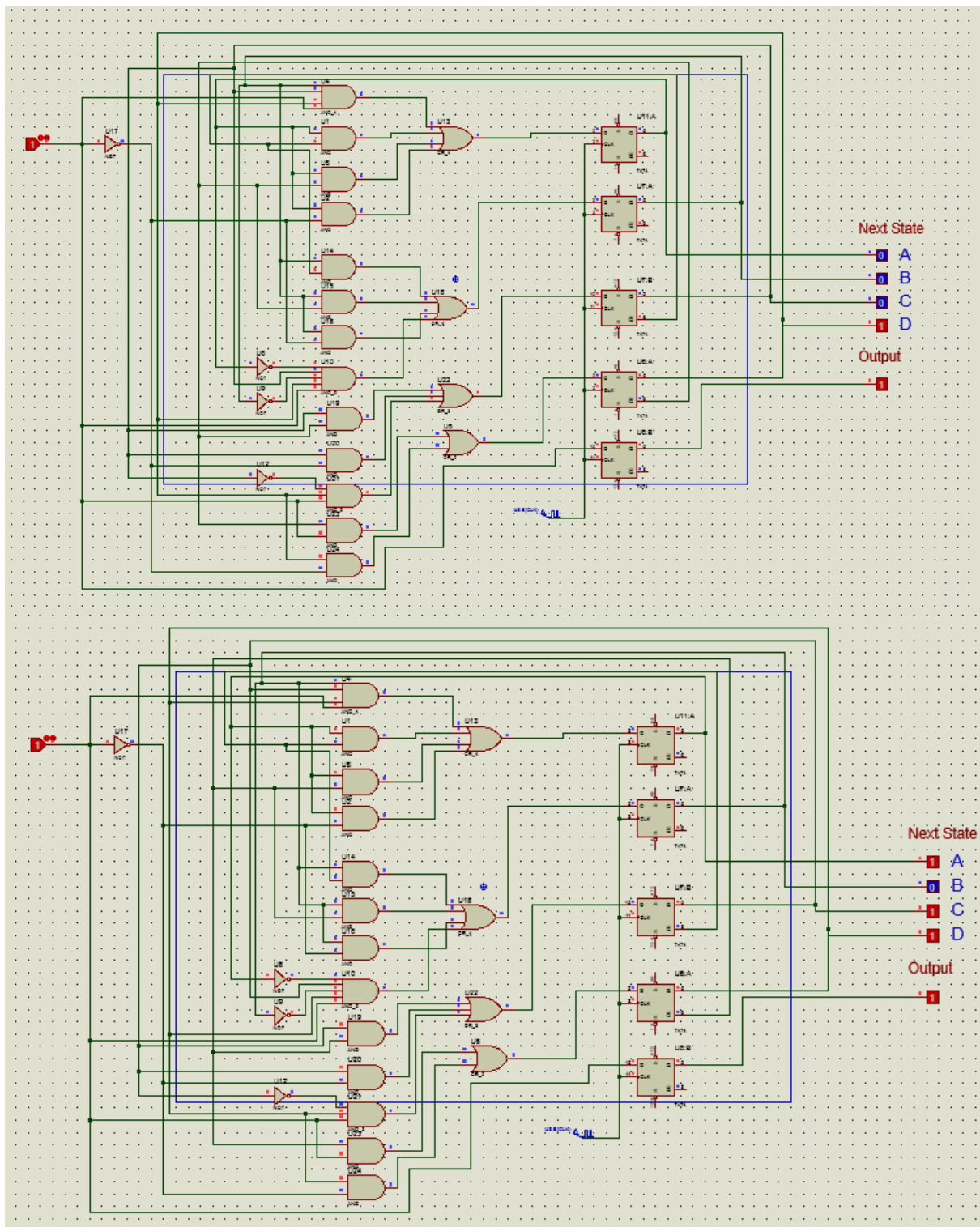
D_4	$\bar{D}\bar{E}$	$\bar{D}E$	$D\bar{E}$	DE
$\bar{A}\bar{B}\bar{C}$	0	1	0	1
$\bar{A}\bar{B}C$	0	1	0	1
$\bar{A}B\bar{C}$	0	1	0	1
$\bar{A}BC$	0	1	0	1
$AB\bar{C}$	0	1	0	1
$A\bar{B}\bar{C}$	0	1	0	1
$AB\bar{C}$	x	x	x	x
ABC	x	x	x	x

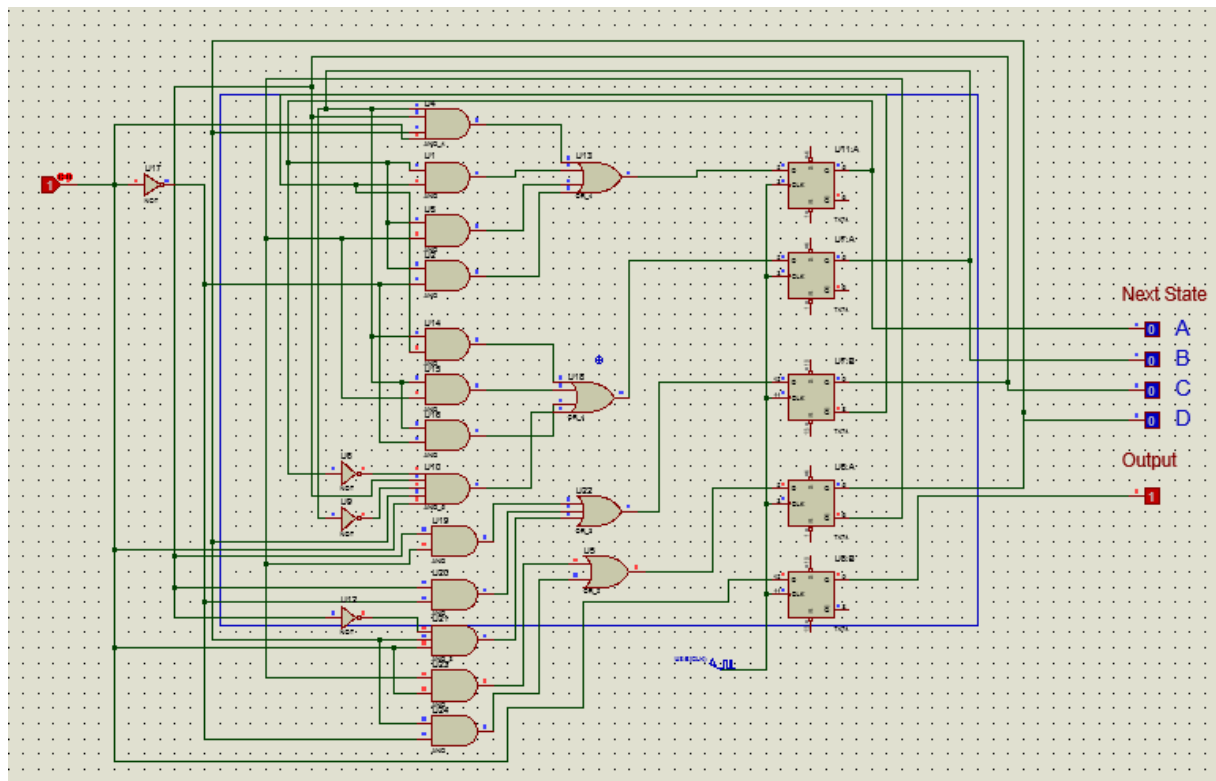
$$D_4 = D'E + DE'$$

$$D_5 = CD' + CE' + C'DE$$

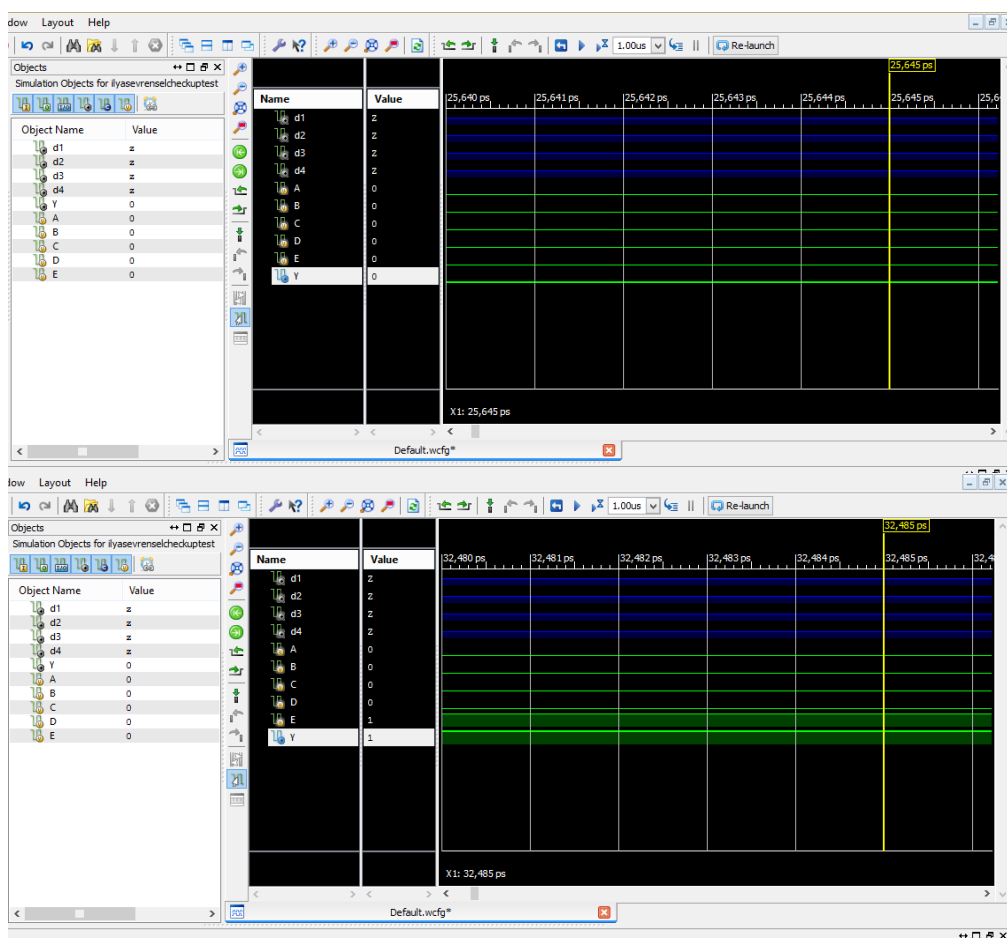
D_5	$\bar{D}\bar{E}$	$\bar{D}E$	$D\bar{E}$	DE
$\bar{A}\bar{B}\bar{C}$	0	1	1	0
$\bar{A}\bar{B}C$	0	1	1	0
$\bar{A}B\bar{C}$	0	1	1	0
$\bar{A}BC$	0	1	1	0
$AB\bar{C}$	0	1	1	0
$A\bar{B}\bar{C}$	0	1	1	0
$AB\bar{C}$	x	x	x	x
ABC	x	x	x	x

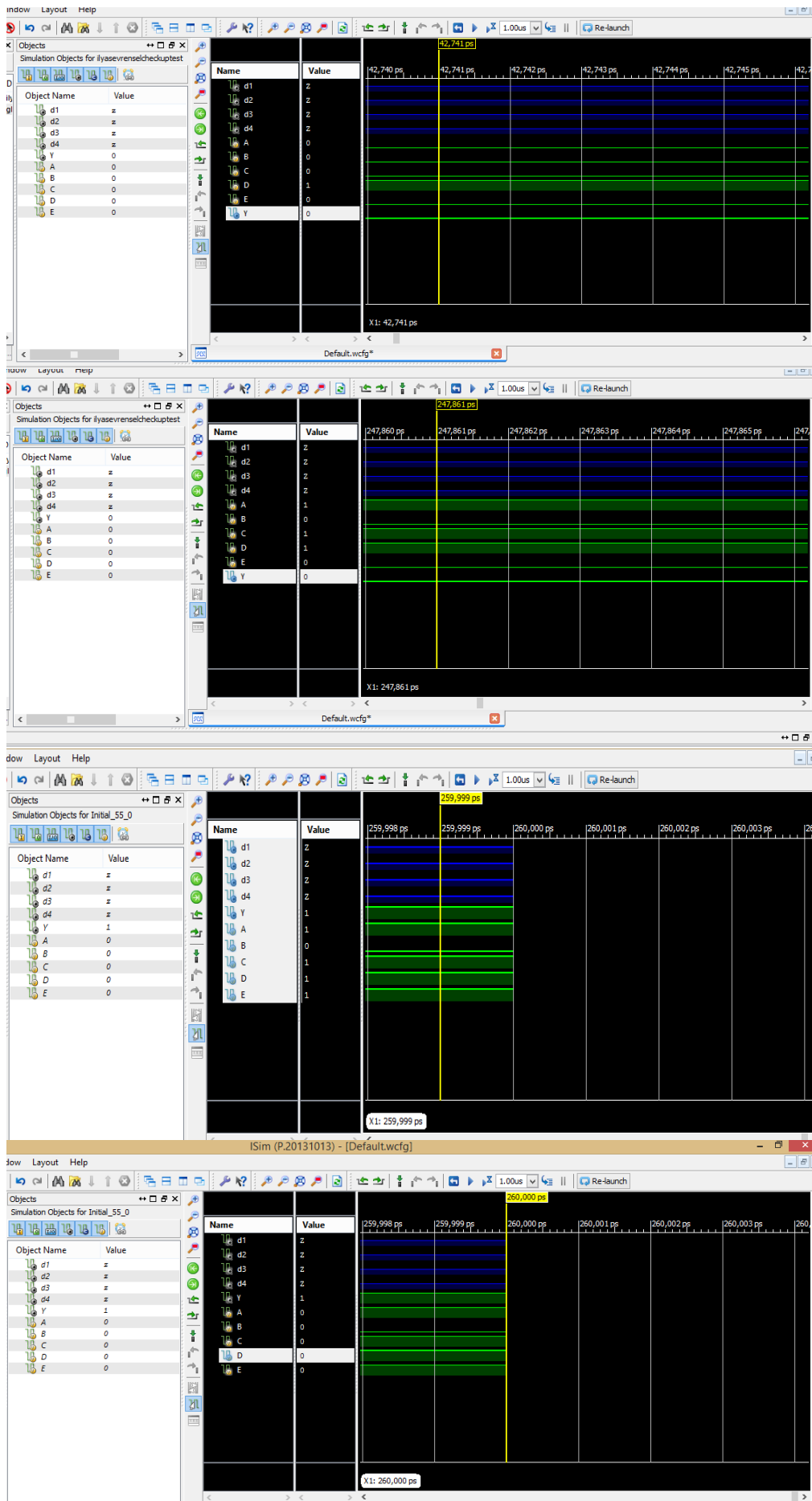
$$D_5 = E$$





4-Xilinx And Codes






```

`timescale 1ns / 1ps
/////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 17:47:04 012/06/2022
// Design Name:
// Module Name: mustafayagcicheckup
// Project Name:
// Target Devices:
// Tool versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
/////////////////////////////////////////////////////////////////
module mustafayagcicheckup( A,B,C,D,E,d1,d2,d3,d4,Y
);
    input A,B,C,D,E;
    output d1,d2,d3,d4,Y;
    wire k,l,m,n,o,w1,w2,w3,w4,w5,w6,w7,w8,w9,w10,w11,w12,w13,w14,w15,w16;

    assign Y=E;
    not g1(k,A);
    not g2(l,B);
    not g3(m,C);
    not g4(n,D);
    not g5(o,E);

    and g6(w1,A,m);
    and g7(w2,A,n);
    and g8(w3,A,o);
    and g9(w4,B,C,D,E);

    and g10(w5,B,m);
    and g11(w6,B,n);
    and g12(w7,B,o);
    and g13(w8,k,l,C,D,E);

    and g14(w9,C,n);
    and g15(w10,C,o);
    and g16(w11,D,E,m);

    and g17(w12,E,n);
    and g18(w13,D,o);

    or g19(w14,w1,w2,w3,w4);
    or g20(w15,w5,w6,w7,w8);
    or g21(w16,w9,w10,w11);
    or g22(w17,w12,w13);

endmodule

```

```

`timescale 1ns / 1ps

```

```

////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 17:47:04 012/06/2022
// Design Name: mustafayagcicheckup
// Module Name: D:/new/mustafayagcicheckup/mustafayagcicheckuptest.v
// Project Name: mustafayagcicheckup
// Target Device:
// Tool versions:
// Description:
//
// Verilog Test Fixture created by ISE for module: mustafayagcicheckup
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
////////////////////////////////////

```

```

module mustafayagcicheckuptest;

    // Inputs
    reg A;
    reg B;
    reg C;
    reg D;
    reg E;

    // Outputs
    wire d1;
    wire d2;
    wire d3;
    wire d4;
    wire Y;

    // Instantiate the Unit Under Test (UUT)
    mustafayagcicheckup uut (
        .A(A),
        .B(B),
        .C(C),
        .D(D),
        .E(E),
        .d1(d1),
        .d2(d2),
        .d3(d3),
        .d4(d4),
        .Y(Y)
    );

    initial begin
        // Initialize Inputs
        A = 0;
        B = 0;
        C = 0;
        D = 0;
        E = 0;

        #20; //delay wait for running

        A = 1'b0; B=1'b0;C=1'b0;D=1'b0;E=1'b0;
        #10;
        A = 1'b0; B=1'b0;C=1'b0;D=1'b0;E=1'b1;
    end

```

```
#10;
A = 1'b0; B=1'b0;C=1'b0;D=1'b1;E=1'b0;
#10;
A = 1'b0; B=1'b0;C=1'b0;D=1'b1;E=1'b1;
#10;
```

```
A = 1'b0; B=1'b0;C=1'b1;D=1'b0;E=1'b0;
#10;
A = 1'b0; B=1'b0;C=1'b1;D=1'b0;E=1'b1;
#10;
A = 1'b0; B=1'b0;C=1'b1;D=1'b1;E=1'b0;
#10;
A = 1'b0; B=1'b0;C=1'b1;D=1'b1;E=1'b1;
#10;
```

```
A = 1'b0; B=1'b1;C=1'b0;D=1'b0;E=1'b0;
#10;
A = 1'b0; B=1'b1;C=1'b0;D=1'b0;E=1'b1;
#10;
A = 1'b0; B=1'b1;C=1'b0;D=1'b1;E=1'b0;
#10;
A = 1'b0; B=1'b1;C=1'b0;D=1'b1;E=1'b1;
#10;
```

```
A = 1'b0; B=1'b1;C=1'b1;D=1'b0;E=1'b0;
#10;
A = 1'b0; B=1'b1;C=1'b1;D=1'b0;E=1'b1;
#10;
A = 1'b0; B=1'b1;C=1'b1;D=1'b1;E=1'b0;
#10;
A = 1'b0; B=1'b1;C=1'b1;D=1'b1;E=1'b1;
#10;
```

```
A = 1'b1; B=1'b0;C=1'b0;D=1'b0;E=1'b0;
#10;
A = 1'b1; B=1'b0;C=1'b0;D=1'b0;E=1'b1;
#10;
A = 1'b1; B=1'b0;C=1'b0;D=1'b1;E=1'b0;
#10;
A = 1'b1; B=1'b0;C=1'b0;D=1'b1;E=1'b1;
#10;
```

```
A = 1'b1; B=1'b0;C=1'b1;D=1'b0;E=1'b0;
#10;
A = 1'b1; B=1'b0;C=1'b1;D=1'b0;E=1'b1;
#10;
A = 1'b1; B=1'b0;C=1'b1;D=1'b1;E=1'b0;
#10;
A = 1'b1; B=1'b0;C=1'b1;D=1'b1;E=1'b1;
#10;
A = 1'b0; B=1'b0;C=1'b0;D=1'b0;E=1'b0;
```

```
$finish;
```

```
end
```

```
endmodule
```

DISCUSSION AND CONCLUSIONS

I tried to use all the information I had learned since the beginning of the year when I was doing this project. I made a lot of use of the experiment 8 experiment, especially when writing my code. I wrote my circuit in the style of gate level modeling, but I learned how to write dataflow modeling. In this experiment, I especially understood the two forms of writing and the working logic of the Xilinx program better.

I took a screenshot from different points when I printed it out. My Xilinx simulation outputs exactly match the state table I have made. When I drew my project in proteus, I compared my results and saw the results I wanted.

As a result, I learned the Xilinx program at the entry level, and I think this project has contributed a lot.